CS 532, Spring 2019: Quiz 9

NAME:

Problem 1. Consider a 3D plane $\pi = (A, B, C, D)^\top$, a pinhole camera with focal length f located at the 3D origin with orientation equal to $\mathbf{I}$, and a pixel $\mathbf{p}$ with image plane coordinates $(x', y')^\top$. What is the 3D depth of the intersection between $\mathbf{p}$'s viewing ray and the 3D plane π? Make an illustrative sketch.

Hint: You can leverage the plane-point intersection equation $\pi \cdot (X, Y, Z, 1) = 0$, as well as the similar triangle equality (e.g. $\frac{x'}{f} = \frac{X}{Z}$) applicable to pinhole cameras.